

Muhammmad Athar Abbas
SP25-BSE-082
Object Oriented Programming
Lab Assignment 4

Task 1:

```
package application.labtask;

import javafx.event.ActionEvent;
import javafx.event.EventHandler;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.RadioButton;
import javafx.scene.control.Separator;
import javafx.scene.control.ToggleGroup;
import javafx.scene.layout.FlowPane;
import javafx.stage.Stage;
import javafx.application.Application;

public class Task1 extends Application {

    public static void main(String[] args) {
        launch(args);
    }

    @Override
    public void start(Stage stage) throws Exception{

        Label choose = new Label("Select your semester");
        Label response = new Label("No semester mentioned.");
        Button btnConfirm = new Button("Confirm Semester Selection");
        RadioButton rbOne = new RadioButton("One");
        RadioButton rbTwo = new RadioButton("Two");
        RadioButton rbThree = new RadioButton("Three");
        ToggleGroup tg = new ToggleGroup();
        rbOne.setToggleGroup(tg);
```

```
        rbTwo.setToggleGroup(tg);
        rbThree.setToggleGroup(tg);
        rbOne.setSelected(true);
        Separator separator = new Separator();
        separator.setPrefWidth(190);

        stage.setTitle("Lab JavaFX");

        FlowPane rootNode = new FlowPane(10,10);
        rootNode.setAlignment(Pos.CENTER);

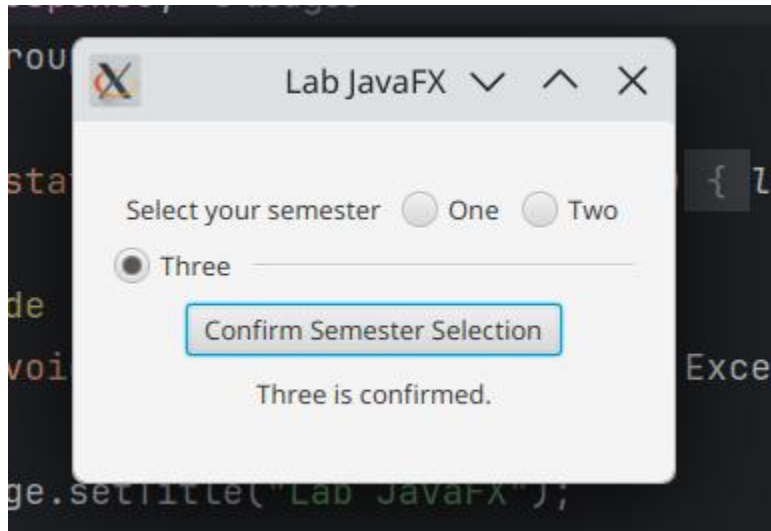
        btnConfirm.setOnAction(new EventHandler<ActionEvent>() {
            @Override
            public void handle(ActionEvent event) {
                RadioButton rb = (RadioButton) tg.getSelectedToggle();
                response.setText(rb.getText() + " is confirmed.");
            }
        });
        rootNode.getChildren().addAll(choose, rbOne, rbTwo, rbThree,
separator, btnConfirm, response);

        Scene myScene = new Scene(rootNode, 300, 180);
        stage.setScene(myScene);
        stage.show();
        stage.centerOnScreen();

    }

}
```

Output:



Task 2:

```
package application.labtask;

import javafx.application.Application;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.fxml.FXMLLoader;
import javafx.scene.Parent;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.stage.Stage;

public class Task2 extends Application {

    @FXML
    private Button btnConfirm;
    @FXML
    private ToggleGroup tg;
    @FXML
    private Label response;
```

```

public static void main(String[] args){
    launch(args);
}
@Override
public void start(Stage stage) throws Exception {

    stage.setTitle("Lab JavaFX from FXML");
    FXMLLoader loader = new
FXMLLoader(getClass().getResource("MyLayout.fxml"));
    Parent root = loader.load();
    Scene scene = new Scene(root);
    stage.setScene(scene);
    stage.show();
    stage.centerOnScreen();

}

public void onClickHandler(ActionEvent event){
    RadioButton rb = (RadioButton) tg.getSelectedToggle();
    response.setText(rb.getText() + " is confirmed.");
}

}

```

FXML:

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.scene.control.*?>
<?import javafx.scene.layout.FlowPane?>

<FlowPane xmlns="http://javafx.com/javafx/8"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="application.labtask.Task2" alignment="CENTER" hgap="10"
vgap="10" prefWidth="300" prefHeight="180">

    <fx:define>

```

```
<ToggleGroup fx:id="tg"/>
</fx:define>
<Label text="Select your semester"/>

<RadioButton text="One" toggleGroup="$tg" selected="true"/>
<RadioButton text="Two" toggleGroup="$tg"/>
<RadioButton text="Three" toggleGroup="$tg"/>

<Separator prefWidth="190"/>

<Button fx:id="btnConfirm" text="Confirm Semester Selection"
onAction="#onClickHandler"/>

<Label fx:id="response" text="No semester mentioned."/>
</FlowPane>
```

Output:

